

N0JCG NOAA Weather Radio

Operator User Guide | Preview release v0.1.1

A focused receive-only Raspberry Pi appliance for all seven US NOAA Weather Radio channels, FFT-directed strongest-channel selection, NFM audio, and operator-configurable SAME alert filters.

Safety and operating boundary

This product has no transmit path. It is not an emergency alert replacement; keep an official weather receiver or other warning source available. A green application status proves software state, not RF coverage or the correctness of a warning.

Hardware and identity

Use a Raspberry Pi with current 64-bit Raspberry Pi OS, stable power, network access, an RTL-SDR with EEPROM serial **00000162**, and a suitable VHF antenna. USB enumeration is not serial ownership; the application passes the required serial to RTL tools and does not use a temporary USB index.

Installation

1. From an MSYS2 UCRT64 shell, run `./deploy/install.sh --check-only`.
2. Run `./deploy/install.sh` on the target Pi.
3. Open `http://<pi-address>:8086/` and confirm serial 00000162.
4. Use `systemctl status` and `journalctl` for service evidence.

First operation

1. Confirm the receiver serial card shows 00000162.
2. Press Start. The application surveys 162.395-162.555 MHz and starts browser audio on the strongest valid candidate.
3. Review candidate cards; click any card to tune directly to that NOAA channel.
4. Press Stop—the same button changes from Start to Stop—before disconnecting the receiver or changing USB hardware.

Simulation mode selects a deterministic test winner for software validation and is not live RF proof.

FFT scan and audio

The scanner scores power around each canonical channel from one wide survey. The highest SNR candidate is selected, then `rtl_fm` is started in narrow-FM mode at 48 kHz. The server supplies short 24 kHz mono WAV segments to the browser for scheduled playback. A high peak can still be interference; verify intelligible NOAA audio.

SAME alerts

Enter county FIPS codes and SAME event codes such as TOR or SVR, separated by commas. Empty fields accept all values. The test parser exercises syntax only; it does not prove that an over-the-air warning was received.

Record the raw header and UTC receipt time for live validation. Verify the tuned channel and intelligible 1050 Hz alert sequence.

Registration

The product has its own registration namespace, n0jcg-noaa-weather-radio. The local registration endpoint reports a unique installation identifier and trial or registered mode. Activation tokens are product-scoped and stored in local runtime state.

Troubleshooting

- No device: check rtl_test -d 00000162, USB power, permissions, and competing SDR owners.
- No candidates: check antenna, gain, local NOAA coverage, and rtl_power installation.
- Wrong winner: inspect all SNR values and reduce gain if the receiver is saturated.
- No SAME alert: validate with a live or recorded SAME fixture.
- Shared receiver conflict: stop Air Traffic Center VHF audio before starting a live scan here.

Release boundary

v0.1.1 includes software tests, deterministic simulation, compact operator UI, direct channel tuning, scheduled browser WAV audio, FFT scoring, NFM process control, SAME parsing, registration state, installer, package tooling, and this guide. Live USB, RF audio, antenna coverage, and end-to-end SAME acceptance remain target-hardware gates.

Channel	Frequency
WX1	162.400 MHz
WX2	162.425 MHz
WX3	162.450 MHz
WX4	162.475 MHz
WX5	162.500 MHz
WX6	162.525 MHz
WX7	162.550 MHz

NOAA channel plan